

```

1  # Note 17: Data Cleaning Methods [MA, GF & MF]
2  # =====
3
4  # Import Libraries
5  import numpy as np
6  from scipy.signal import medfilt
7  from scipy.ndimage import gaussian_filter1d
8
9  # 01. 7MA Code
10 # =====
11
12 # Define X values
13 X = np.arange(1, 23)
14
15 # Define Y values
16 Y_AD1 = np.array([
17     10, 12, 15, 14, 18, 20, 22,
18     19, 21, 25, 24, 28, 30, 27,
19     29, 32, 35, 33, 36, 38, 40, 42
20 ])
21
22 # Prepare X Values for MA-7
23 X_MA7 = X[0:-6]
24
25 # Calculate 7-point Moving Average
26 Y_MD1 = np.convolve(Y_AD1, np.ones(7), 'valid') / 7
27
28 # Print Results
29 print("\nX Values:", X_MA7)
30 print("MA-7 Values:", Y_MD1)
31
32 # 02. Median Filter
33 # =====
34
35 # Numerical data
36 data = np.array([10, 12, 100, 14, 15, 13, 200, 16, 18])
37
38 # Apply Median Filter
39 filtered_data = medfilt(data, kernel_size=3)
40
41 # Print results
42 print("\nOriginal Data:")
43 print(data)
44
45 print("Filtered Data:")
46 print(filtered_data)
47
48 # 03. Gaussian Filter
49 # =====
50
51 # Numerical data
52 data = np.array([10, 12, 15, 100, 14, 16, 18, 200, 20])
53
54 # Apply Gaussian Filter
55 filtered_data = gaussian_filter1d(data, sigma=1)
56
57 # Print results
58 print("\nOriginal Data:")
59 print(data)
60
61 print("Gaussian Filtered Data:")
62 print(filtered_data)

```